Customer Shopping Behavior Analysis

1. Project Overview

The Customer Shopping Behaviour Analysis project is a data-driven initiative designed to uncover insights into consumer purchasing patterns, segment customers into meaningful groups, and identify seasonal shopping trends; using transaction datasets processed through Python-based tools in Jupyter Notebook, the project applies data cleaning, exploratory analysis, clustering, and predictive modelling to highlight key segments such as loyal buyers, bargain hunters, premium shoppers, and occasional buyers, while also revealing cyclical demand peaks during holidays and discount periods, ultimately providing actionable recommendations for marketing strategies, product positioning, customer retention programs, and inventory management.

2. Dataset Summary

Rows: 3,900

Columns: 18

Key Features:

- **Customer Information:** Unique customer IDs, demographic attributes (age, gender, location).
- **Transaction Records:** Purchase dates, product categories, quantities, and basket values.
- **Behavioural Attributes:** Frequency of purchases, average spend, discount usage.
- **Temporal Data:** Seasonal and monthly purchase patterns.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python.

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Used df.info () to check the structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check**: Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed the structured analysis in PostgreSQL to answer key business questions.

1. **Revenue by Gender:** Compare total revenue generated by male versus female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **Discount Usage and High Spending:** Find customers who used a discount but still spent more than the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81

3. **Top 5 Products by Review Rating:** List the five products with the highest average customer review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Purchase Amounts by Shipping Type:** Compare average purchase amounts between standard and express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscriber vs Non-Subscriber Spending:** Compare average spend and total revenue between subscribed and non-subscribed customers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Products with Highest Discount Usage:** Identify the five products with the highest percentage of purchases made with discounts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation by Purchase History:** Segment customers into **New**, **Returning**, and **Loyal** based on previous purchases.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category:** Determine the three most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers and Subscription Likelihood:** Analyse whether customers with more than five previous purchases are likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue Distribution by Age Group:** Show total revenue contribution by different age groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-ages	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

6. Business Recommendations

- Develop **personalized marketing campaigns** tailored to each segment.
- Enhance **premium product offerings** with exclusivity.
- Build **real-time dashboards** for monitoring customer trends.
- Enhance **personalization** in online shopping experiences.